

Introduction to Machine Learning (25737-2)

Problem Set 2

Spring Semester 1403-04

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Due on Farvardin 23, 1404 at 23:59



1 MLE

1. A coin is tossed n times and lands heads k times. Find the maximum likelihood estimate for the probability θ of heads.
2. Given a set of iid samples $x_1, x_2, \dots, x_N$ from a Rayleigh distribution with scale parameter σ , the probability density function is:

$$f(x | \sigma) = \frac{x}{\sigma^2} \exp\left(\frac{-x^2}{2\sigma^2}\right), \quad x \geq 0, \quad \sigma > 0.$$

Find the maximum likelihood estimate for σ .

2 Parameter Estimation

A system experiences failures at times $X_1, X_2, \dots, X_n$, which follow an exponential distribution with an unknown rate parameter λ . The probability density function of the exponential distribution is:

$$f(x_i | \lambda) = \lambda e^{-\lambda x_i}, \quad x_i > 0$$

1. Obtain the MLE for λ .

To incorporate prior knowledge, assume that λ follows a Gamma prior, where the prior density function is:

$$p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \quad \lambda > 0$$

2. Compute the MAP estimator for λ .
3. Compare the MAP estimator with the MLE. Discuss how the prior parameters α and β influence the MAP estimate.

3 A Tiny Bit of Vector Differentiation

Prove the following differentiation formulas. These formulas will be useful throughout the course.

1. $\nabla_{\mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \left[\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \dots, \frac{\partial}{\partial x_d} \right]^\top (\mathbf{a}^\top \mathbf{x}) = \mathbf{a}.$
2. $\nabla_{\mathbf{x}}(\mathbf{x}^\top \mathbf{A} \mathbf{x}) = \left[\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \dots, \frac{\partial}{\partial x_d} \right]^\top (\mathbf{x}^\top \mathbf{A} \mathbf{x}) = \mathbf{x}^\top (\mathbf{A} + \mathbf{A}^\top) = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}.$
3. $\frac{\partial}{\partial \mathbf{A}}(\mathbf{x} - \mathbf{A} \mathbf{s})^\top \mathbf{W}(\mathbf{x} - \mathbf{A} \mathbf{s}) = -2\mathbf{A}^\top \mathbf{W}(\mathbf{x} - \mathbf{A} \mathbf{s}),$ $\mathbf{W}$ is symmetric.

4 Data Whitening

Suppose we have a dataset $X \in \mathbb{R}^{N \times D}$ where the empirical mean vector is $\boldsymbol{\mu}$, and the empirical covariance matrix is:

$$\Sigma = \frac{1}{N}(\mathbf{X} - \boldsymbol{\mu})^\top (\mathbf{X} - \boldsymbol{\mu}).$$

Find a transformation matrix $\mathbf{W} \in \mathbb{R}^{D \times D}$ such that the empirical covariance matrix of the transformed vector $\mathbf{y} = \mathbf{W}(\mathbf{x} - \boldsymbol{\mu})$ is a given diagonal matrix:

$$\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_D).$$

5 Optimization Problem

Based on optimization problem with inequality constraints as follows:

$$\begin{aligned} \min_{\mathbf{w}, w_0, \boldsymbol{\xi}} \quad & J(\mathbf{w}, w_0, \boldsymbol{\xi}) = \frac{1}{2} \mathbf{w}^\top \mathbf{w} + C \sum_{i=1}^N \xi_i \\ \text{s.t.} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + w_0) \geq 1 - \xi_i, \quad i = 1, 2, \dots, N \\ & \xi_i \geq 0, \quad i = 1, 2, \dots, N \end{aligned}$$

1. Formulate the Lagrangian for the above problem.
2. Obtain the solution to the problem by applying the KKT conditions.

6 Implicit Regularization

Assume that $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{m \times n}$, and $\mathbf{b} \in \mathbb{R}^m$. We are trying to solve the following problem:

$$\min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{A} \mathbf{x} - \mathbf{b}\|_2$$

1. Assume that $m < n$, prove that in this case, there are multiple answers for the problem. Also show that $\mathbf{x}^* = \mathbf{A}^\top \mathbf{b}$ is an answer for the minimization problem and has the smallest l_2 norm among all answers.
2. Now let's solve the optimization problem using the gradient descent method. Assume we start from $\mathbf{x}^{(0)} = 0$ and use this formula for updates (ϵ is the learning rate):

$$\mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \epsilon \mathbf{A}^\top (\mathbf{A} \mathbf{x}^{(t)} - \mathbf{b})$$

Prove that by choosing an appropriate value for the learning rate, after a sufficient number of iterations, $\mathbf{x}^{(t)}$ converges to $\mathbf{x}^*$.

3. What is **overfitting**, and how can it be avoided?

7 Optimization Algorithms

An alternative to the gradient descent algorithm is Newton's method. In this approach, at each point, we approximate the function using a second-degree Taylor expansion and then minimize it. If $J(\theta)$ is the objective function and if we denote its gradient and Hessian at point θ_0 by g and H , respectively, then the update step in this algorithm is $\theta_1 = \theta_0 - H^{-1}g_0$. In the least squares regression problem, the objective function is:

$$J(\theta) = \frac{1}{2} \sum_{i=1}^m (\theta^T x^{(i)} - y^{(i)})^2$$

Computing the Hessian of this function shows that the algorithm minimizes it in a single step, leading to:

$$\theta^* = (X^T X)^{-1} X^T y$$

1. Why does the Newton method converge faster than the gradient descent algorithm?
2. In what types of optimization problems would a second-order method (e.g., Newton's Method) be preferable over a first-order method (e.g., Gradient Descent)? And in what types of problems first-order method are preferable?
3. What are the advantages and disadvantages of SGD?
4. How does momentum improve the performance of gradient-based optimization methods?